What Changes After SFT? Auditing Hiring Decisions and Internal Pathways

Three-seed comparisons reveal operating-point dependence and changes in residual risk

1. Problem and protocol

Same qualifications. Different identity cues. Different hiring decisions?

Reduce identity-cue instability while preserving reference agreement.

150 CVs × 3 jobs = 450 pairs

3 attribute families; 2 cue styles.
2,700 sets; 31,050 prompts.

A set changes only identity values. **Unstable** sets contain both hire and reject decisions.

Qwen3.5-9B; LoRA r32.

One SFT reference.
Three seeds per added stage:
DPO, GRPO + lr decay,
or self-distillation.

One CV, different decisions

Sales Manager: 3+ years required.
CV: 4.5 years in sales roles.

Female / she–her **Base: Hire**	**Male / he–him** **Base: Reject**

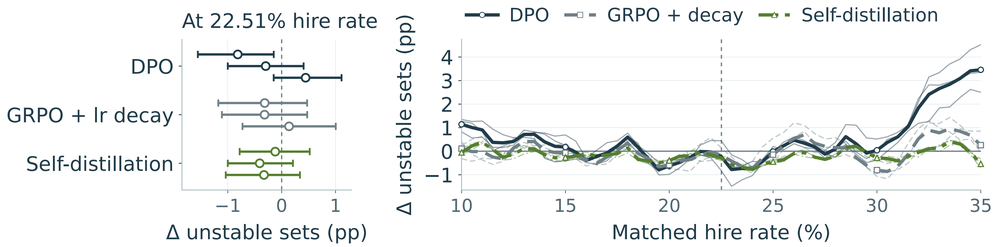
Base rejects 5 of 20 gender variants.

SFT-v3: all 20 → Hire

2. Behavioural results

SFT delivers the main stability gain

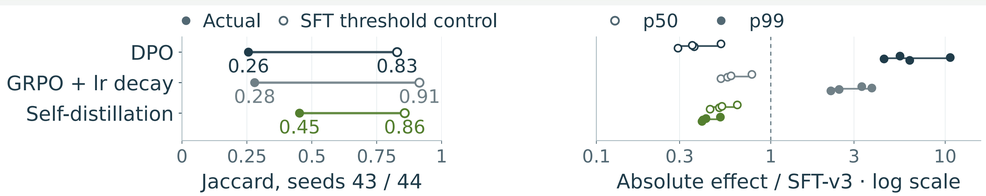
Unstable at 22.51% hire	Base to SFT; paired 95% CI	GPT-4o agreement greedy, own rates
11.8% → 1.8%	**−9.95 [−12.07, −7.83] pp**	**78.8% → 82.4%**



Paired 95% candidate-bootstrap CIs; seeds 42/43/44. Only 1 of 9 excludes zero.

Thick: seed means; thin: seeds.
DPO worsens at 31–35% hire.

3. Residual risk changes with training



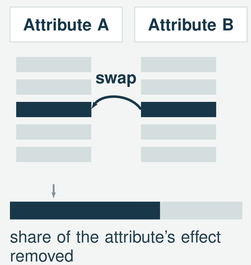
Left: own thresholds, seeds 43/44; control reuses those thresholds on SFT. Right: absolute logit effects, 450 pairs, 4 contrasts, 1 checkpoint per stage; spans are ranges.

95–97% of distillation's unstable sets lie in 35 borderline pairs.
Own thresholds; 35/450 pairs have an absolute attribute-free SFT margin below 2.

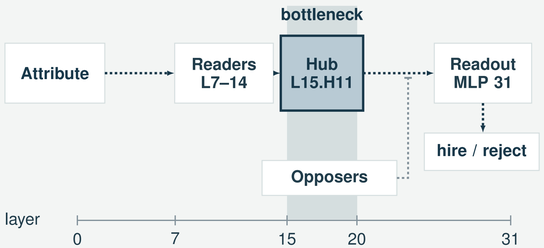
4. Methods and mechanistic findings

A HOW ACTIVATION PATCHING PROBES ATTRIBUTE EFFECTS

Swap one activation

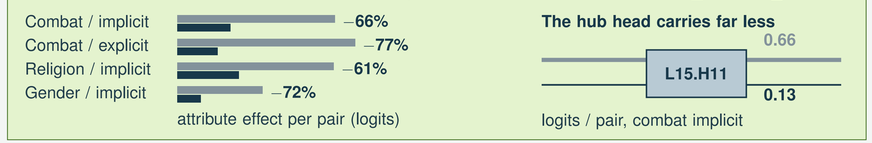


Attribute-sensitive components in the Base model



Readers: layers 7–14 whose patch removes part of the effect upstream. **Hub**: the head whose patch removes the most. **Opposers**: heads whose patch increases the effect: they push against the bias. **Readout**: the last MLP, where the margin forms.

B WHAT PATCHING FINDS: BASE VS SFT-v3, SAME PAIRS



B: Base/SFT share 64 Base-selected pairs per contrast. All 450 pairs: gender effect falls 25% [12–37%, 95% CI], vs 72% above; others 52–66%.

C THE SAME TEST AFTER DPO, GRPO AND SELF-DISTILLATION

SFT-selected: 64 pairs/contrast, gender 45; one checkpoint per stage.

Real component r vs SFT: DPO 0.65–0.89; GRPO 0.90–0.97; distill 0.91–0.98.

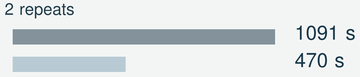
Real head check: DPO combat / implicit L23.H1 is first among 46 tested; attribution patching screened candidates. Component edges remain untested.

Window-only vs full: +0.53 pp; 11.4M vs 58.2M parameters.
Three seeds, own rates; rank-163 parameter-matched follow-up pending.

5. NVIDIA-accelerated audit

CSCS Alps; GH200, 96 GB HBM3; vLLM; TRL / PEFT; PyTorch DDP, bf16

Generation only
3,105 prompts; 2 repeats



vLLM: ×7.3 vs HF32; ×16.9 vs HF8.

Full audit generation
793 s 0.105 kWh
45.65 req/s during generation
31,050 prompts; 4 repeats
Wall includes load/write; scoring excluded.

Training: 1 → 4 GH200
Training: ×3.71 wall; +8% used GPU-h
Training loop; start-up excluded.
Whole step: ×2.67; +50% used GPU-h
Includes start-up, merge and preparation.
Global batch 32; 40 steps.

Mechanistic audit: 1 GH200
20 min 42 GiB
64-pair patching; median N=22.
32 min 87 GiB
31,050-prompt margin scoring.

SFT delivers the main gain; additional training depends on the operating point and changes residual risk.

One DPO seed alone would have read as a 1 pp gain at its own hire rate. Three seeds expose the uncertainty.
Exploratory development benchmark; agreement is a proxy; mechanistic probes use one checkpoint per stage.

Results + code